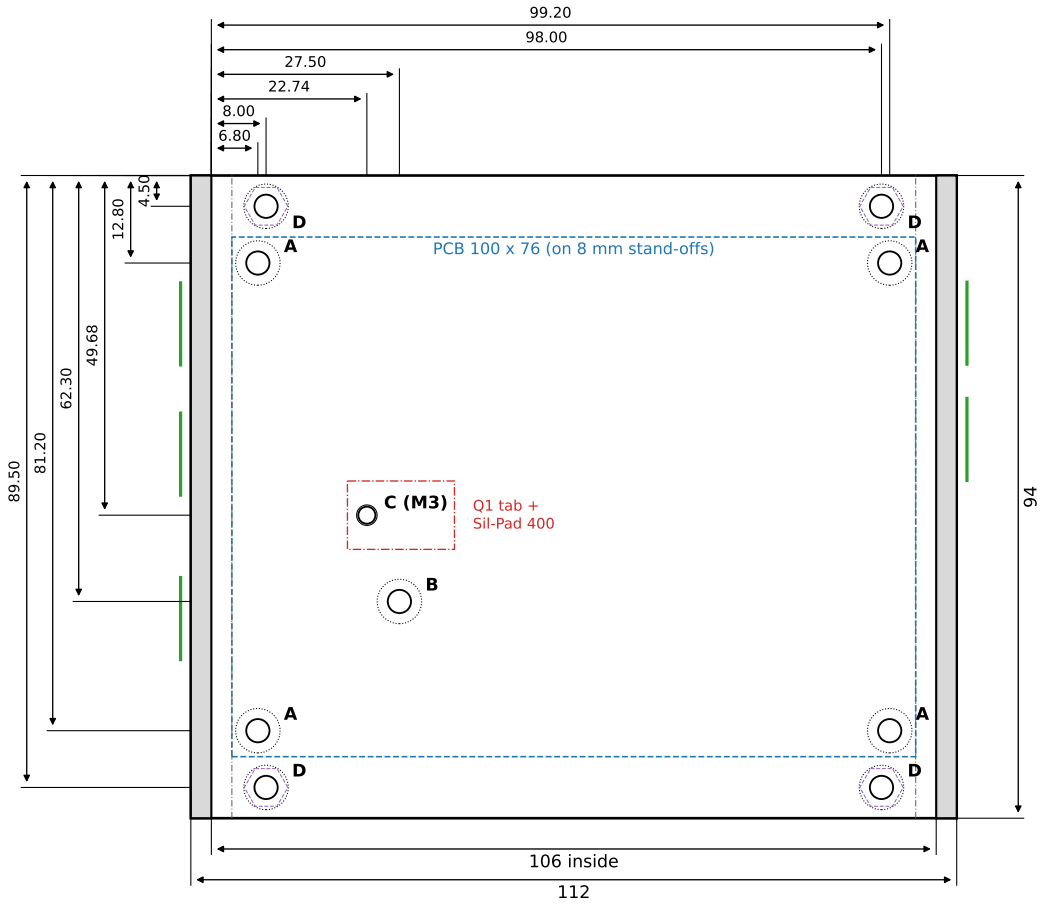
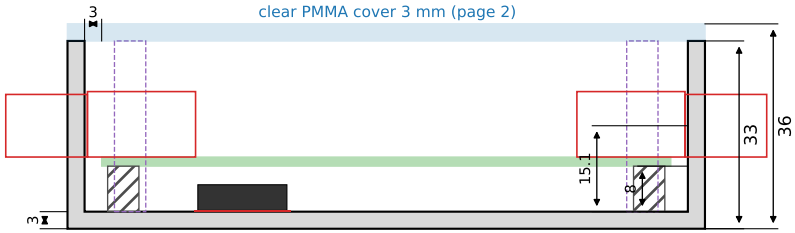
TOP VIEW (looking down on the floor; PCB outline dashed)



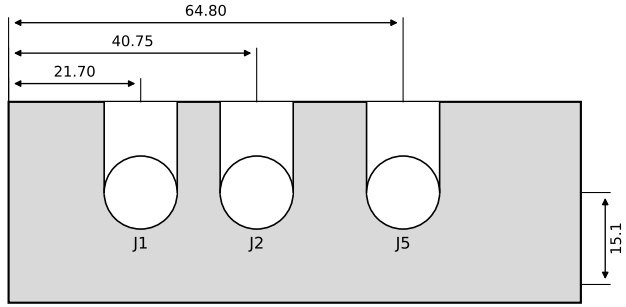
x from the INSIDE face of the left wall, y from the UPPER floor edge in this view (PCB orientation as in KiCad).

SIDE VIEW (from the upper edge), PCB and sockets thin



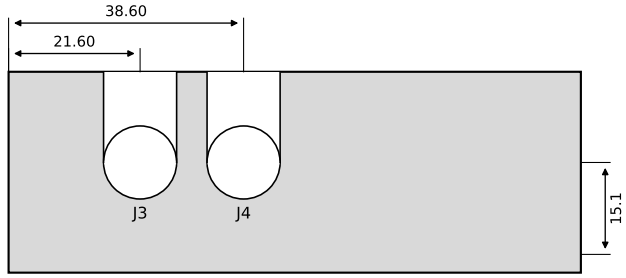
bends: inside R3, 90 deg; walls and floor 3 mm

LEFT WALL, seen from INSIDE (J1 VIN+, J2 VIN-, J5 CHASSIS)



y from the upper floor edge of the top view; slots 12 wide, open top, R6 (centre z = 15.1)

RIGHT WALL, seen from OUTSIDE (J3 VOUT+, J4 VOUT-)



y from the upper floor edge of the top view; slots 12 wide, open top, R6 (centre z = 15.1)

HOLE TABLE

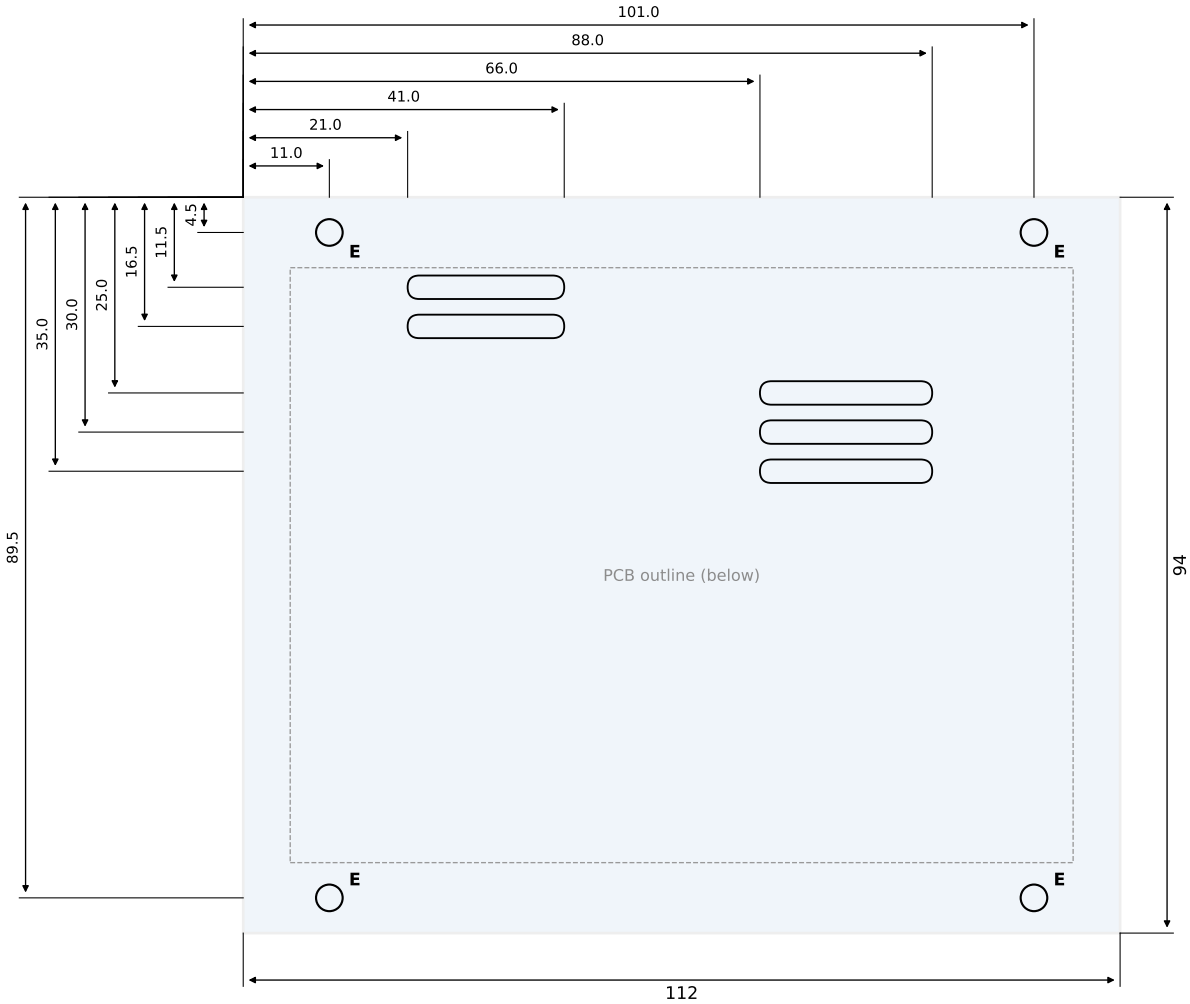
A	4	D3.4 through, countersunk 90 deg to D6.5 from below	M3x6 csk screw from below, M3x8 NYLON stand-off (MH1-4)
B	1	D3.4 through, countersunk 90 deg to D6.5 from below	M3x6 csk screw, M3x8 METAL stand-off (MH5 = case contact)
C	1	M3 thread through (tap drill 2.5)	Q1: M3x6 pan screw from above + shoulder washer
D	4	D3.4 through, countersunk 90 deg to D6.5 from below	M3x6 csk screw, M3x30 hex pillar for the cover

NOTES

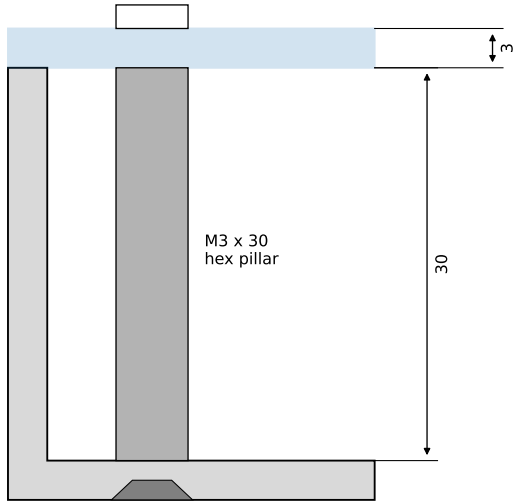
1. Material: aluminium 5754-H22 or 5052-H32, 3 mm sheet (bends well; do not use 6061-T6). Bend inside radius R3.
2. Finish: deburr all holes; the floor under Q1 must be flat and burr-free (Sil-Pad 400 only 0.23 mm).
   Bare/chromated is fine; black anodising runs Q1 and the board ~3-4 K cooler. Mask hole B (metal stand-off contact) and the floor under Q1 flat.
3. Assembly: Sil-Pad + Q1 on the floor; PCB lowered in (socket barrels slide into the open slots); stand-off screws;
   then Q1 screw through the 7 mm PCB hole (torque ~0.6 N m); solder Q1 leads last (leads bent 90 deg at 3.8 mm from the body).
4. Feet: 4 adhesive rubber feet (>= 3 mm) so the case does not touch the reference ground plate when CASE is open.
5. Q1 tab-to-case capacitance through the Sil-Pad ~30 pF: the case is the 'heatsink' of the lecture example.
6. Cover: 3 mm clear PMMA on the wall tops and four M3 x 30 pillars (holes D), 4 x M3 x 8 nylon screws - see page 2.

EMC demo flyback board rev E - aluminium U-channel case (Q1 heatsink; hole C moved in rev E) - all dimensions in mm - scale: fit

TRANSPARENT COVER, 3 mm clear PMMA (acrylic) - top view, same orientation as page 1



SECTION (left wall corner)



x from the LEFT edge of the cover (= outside of the left wall), y from its UPPER edge. Slots 3 mm wide, rounded ends.

NOTES (cover)

1. Material: 3 mm clear cast PMMA (acrylic), laser cut, protective film on both sides until assembly. Polycarbonate (3 mm) is a tougher alternative.

2. Holes E: D3.4 for 4 x M3 x 8 nylon pan screws into the M3 x 30 hex pillars (holes D on page 1). The cover rests on the wall tops (z = 30 mm).

3. Clearance: the tallest part, the CM choke L1, ends at 27.6 mm above the floor (2.4 mm below the cover); the sockets end at 20.6 mm.

4. Ventilation slots over the clamp resistors R9/R10/R20 and over F1/D1 (the hottest parts); air also enters through the open long sides.

5. The cover is an insulator: it does not change the EMC behaviour of the case (CASE still selects the case potential).